

Quantum-Gravity-Inspired Effects in Neutron Stars: A Critical Map from Microscopic Assumptions to Observables

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Abstract

Claims that neutron stars constrain quantum gravity often compare models whose new physics enters at different stages of the stellar calculation. This article provides a critical map of three representative routes: generalized-uncertainty-principle deformations of microscopic thermodynamics, noncommutative-inspired modifications of the effective matter source, and higher-curvature or scale-dependent modifications of gravitational dynamics. For each route we identify the quantity that is changed, the dimensionless parameter controlling the correction, and the consistency conditions required before masses, radii, or tidal deformabilities can be compared with observations. A deformed phase-space measure must be implemented in a thermodynamically consistent interacting equation of state; a smeared density must be completed by a conserved stress-energy tensor, generally with anisotropic pressure; and a gravitational effective field theory must remain within its derivative expansion and be applied consistently to both the equilibrium background and tidal perturbations. Simple scale estimates show that coefficients of natural size suppressed by the Planck scale are negligible in neutron stars. Observable effects consequently constrain model-dependent effective scales or enhanced couplings, not microscopic quantum gravity directly. The resulting framework is intended to separate robust phenomenological bounds from effects produced by inconsistent implementations or by degeneracies with the high-density equation of state.

Keywords: quantum gravity; neutron stars; generalized uncertainty principle; noncommutative geometry; effective field theory; equation of state; modified gravity; multimessenger astronomy

1 Introduction

General Relativity (GR) provides an exceptionally successful description of gravity on macroscopic scales, while quantum mechanics and quantum field theory describe matter

and the nongravitational interactions with remarkable accuracy. Despite their individual success, these frameworks are not expected to remain independent at arbitrarily high energies. A consistent description of gravity in the quantum regime remains one of the central open problems in theoretical physics [2, 3].

The characteristic scale at which quantum-gravitational effects are usually expected to become important follows from the constants $\hbar$, G , and c . The corresponding Planck length, mass, and energy are

$$\ell_{\text{Pl}} = \sqrt{\frac{\hbar G}{c^3}}, \quad M_{\text{Pl}} = \sqrt{\frac{\hbar c}{G}}, \quad (1)$$

$$E_{\text{Pl}} = M_{\text{Pl}} c^2 = \sqrt{\frac{\hbar c^5}{G}}. \quad (2)$$

Here M_{Pl} denotes the unreduced Planck mass. Numerically,

$$\begin{aligned} \ell_{\text{Pl}} &\simeq 1.6 \times 10^{-35} \text{ m}, & M_{\text{Pl}} &\simeq 2.2 \times 10^{-8} \text{ kg}, \\ E_{\text{Pl}} &\simeq 1.2 \times 10^{19} \text{ GeV}. \end{aligned} \quad (3)$$

These scales are far beyond those accessible to present experiments. This separation has motivated a broad program of quantum-gravity phenomenology: rather than attempting to probe the microscopic quantum structure of spacetime directly, one asks whether effective remnants of the underlying theory could survive at much lower energies and leave observable signatures [2, 3].

Compact astrophysical objects provide a natural setting in which to explore this possibility. Neutron stars are particularly interesting because they combine several physical regimes that are difficult to reproduce simultaneously in terrestrial experiments. Their interiors contain strongly interacting matter at densities of several times nuclear saturation density, while their global structure is governed by strong gravitational fields and therefore requires a relativistic description. At the same time, neutron stars are increasingly accessible to observation. Pulsar timing provides precise mass measurements, X-ray pulse-profile modeling constrains stellar masses and radii, and gravitational waves from binary neutron-star mergers probe their tidal response. Neutron stars have consequently become important laboratories for dense nuclear matter, strong-field gravity, and possible departures from their standard descriptions [4–7].

This possibility must nevertheless be interpreted with care. Even the densest regions of neutron stars remain enormously separated from the Planck scale. Observable deviations should therefore not be interpreted as direct manifestations of the microscopic quantum structure of spacetime. The relevant question is instead whether a given quantum-gravity-inspired scenario generates an effective correction that remains sufficiently enhanced at neutron-star energies, densities, or curvature scales to influence macroscopic stellar properties. This point is especially relevant because many phenomenological constructions introduce effective scales or couplings that need not coincide with their naive Planck-scale values.

A useful way of organizing these effects is according to where the new physics enters the stellar-structure problem. Quantum-gravity-inspired corrections can act on the microscopic matter sector, on the effective description of the matter source, or directly on the gravitational dynamics. These possibilities are conceptually distinct, although their observable consequences can overlap.

If the correction acts on the microscopic matter sector, modifications of kinematics, phase space, or statistical mechanics change the thermodynamic quantities entering the equation of state (EOS),

$$P = P(\varepsilon), \quad (4)$$

and therefore modify the stellar configuration even when the gravitational field equations retain their standard GR form. In a second class of models, the localization or stress-energy distribution of the source itself is modified, potentially leading to extended or anisotropic matter configurations. Finally, modifications of the gravitational sector alter the field equations and hence the conditions for hydrostatic equilibrium, even for an otherwise unchanged matter EOS. These different routes from microscopic assumptions to macroscopic observables are summarized schematically in Fig. 1. The separation between matter, source, and gravitational effects provides the organizing principle adopted throughout this article.

An important complication is that these mechanisms do not necessarily produce unique observational signatures. A modification of the EOS can, for example, shift the mass–radius relation in a way that resembles the effect of modified gravitational dynamics. Similar degeneracies can arise in the tidal response. A deviation in a single observable would therefore generally be insufficient to identify its microscopic origin. Meaningful constraints require several observables to be considered together with the uncertainties associated with the high-density EOS.

We examine three representative classes of quantum-gravity-inspired phenomenology. We first consider generalized uncertainty principles (GUPs), focusing on realizations in which the deformation modifies the phase-space measure and, consequently, the thermodynamics of degenerate matter. We then discuss noncommutative-inspired descriptions, where a fundamental limitation on localization can be represented through an effective smearing of the matter source and may lead to modified or anisotropic stress-energy distributions. Finally, we turn to the gravitational sector and discuss higher-curvature corrections from the perspective of gravitational effective field theory (EFT), together with phenomenological scenarios involving scale-dependent gravitational couplings.

This is a focused critical perspective rather than an exhaustive catalogue. Its contribution is twofold. First, it places three frequently conflated mechanisms in one chain from microscopic assumption to observable. Second, it formulates checks that any claimed neutron-star bound must pass: thermodynamic consistency, stress-energy conservation, control of the effective expansion, and a joint treatment of dense-matter uncertainty. A visible numerical change obtained by choosing a large effective parameter is not, by itself, evidence for an amplified Planck-scale correction; it constrains that effective model.

2 Neutron stars as laboratories for quantum-gravity phenomenology

Neutron stars provide a unique environment in which dense nuclear matter and strong gravitational fields coexist. They are the compact remnants of massive stars and typically contain masses of order $1\text{--}2 M_\odot$ within radii of approximately 10–15 km. Their interiors can reach baryon densities of several times the nuclear saturation density,

$$n_0 \simeq 0.16 \text{ fm}^{-3}, \quad (5)$$

placing the stellar core in a regime where the composition and properties of strongly interacting matter remain uncertain [4–7]. This combination of high density, strong gravity, and direct astrophysical access makes neutron stars particularly useful for testing extensions of both dense-matter physics and gravitational theory.

A convenient measure of the strength of the gravitational field is the stellar compactness,

$$C = \frac{GM}{Rc^2}. \quad (6)$$

For a typical neutron star, $C \sim 0.1\text{--}0.3$, so relativistic effects are essential for determining the stellar structure. In GR, a nonrotating neutron star can be described, to a good approximation, as a static and spherically symmetric configuration. The spacetime metric is then written as

$$ds^2 = -e^{2\Phi(r)}c^2dt^2 + \left(1 - \frac{2Gm(r)}{rc^2}\right)^{-1}dr^2 + r^2d\Omega^2, \quad (7)$$

where $m(r)$ is the gravitational mass enclosed within radius r and $\Phi(r)$ determines the temporal component of the metric.

For an isotropic perfect fluid, Einstein’s equations reduce to the Tolman–Oppenheimer–Volkoff (TOV) equations [8, 9],

$$\frac{dm}{dr} = 4\pi r^2 \frac{\epsilon(r)}{c^2}, \quad (8)$$

$$\frac{dP}{dr} = -\frac{G}{c^2} \frac{[\epsilon(r) + P(r)] \left[m(r) + \frac{4\pi r^3 P(r)}{c^2} \right]}{r^2 \left[1 - \frac{2Gm(r)}{rc^2} \right]}, \quad (9)$$

where $\epsilon(r)$ and $P(r)$ denote the local energy density and pressure. The system is closed by specifying an equation of state,

$$P = P(\epsilon). \quad (10)$$

Starting from a central pressure P_c , the equations are integrated outward with $m(0) = 0$ until the pressure vanishes at $r = R$. The total gravitational mass is then $M = m(R)$. Repeating this procedure for different central pressures generates the mass–radius sequence

associated with a given EOS [4, 5].

This construction also makes clear where physics beyond the standard description can enter. A modification of the microscopic matter sector changes the relation $P(\epsilon)$ while leaving the TOV equations unchanged. A modification of gravity instead changes the stellar structure equations themselves. Other scenarios may modify the effective stress-energy tensor and therefore alter the source entering the gravitational equations. Although these possibilities are theoretically distinct, they can lead to similar changes in macroscopic observables. This is one of the main difficulties in using neutron stars to constrain new physics.

The connection with observations has improved substantially in recent years. Precise radio timing has established neutron stars with masses around two solar masses, placing a robust lower bound on the maximum mass supported by any viable stellar model [10–12]. X-ray pulse-profile modeling provides simultaneous information on masses and radii [15–18], while gravitational-wave observations of binary neutron-star mergers probe the tidal response of the stars [23, 24]. These observables test different aspects of the stellar structure and become considerably more informative when analyzed together.

The usefulness of neutron stars for quantum-gravity phenomenology should, however, be understood in the context of the enormous hierarchy between stellar and Planck scales. Typical densities in the inner core are of order

$$\rho_{\text{NS}} \sim 10^{17} - 10^{18} \text{ kg m}^{-3}, \quad (11)$$

whereas the Planck density is

$$\rho_{\text{Pl}} = \frac{c^5}{\hbar G^2} \simeq 5.2 \times 10^{96} \text{ kg m}^{-3}. \quad (12)$$

The ratio is therefore roughly

$$\frac{\rho_{\text{NS}}}{\rho_{\text{Pl}}} \sim 10^{-80} - 10^{-79}. \quad (13)$$

A similarly large separation occurs at the level of characteristic curvature scales. Consequently, neutron stars should not be interpreted as objects approaching the microscopic Planck regime [2, 3].

Their role is instead to test whether a proposed theory or phenomenological model contains corrections that remain relevant far below the Planck scale. Such corrections may alter the microscopic thermodynamics of dense matter, the effective stress-energy tensor, or the gravitational dynamics. If sufficiently large, these changes propagate through the stellar structure problem and can modify quantities such as the maximum mass, radius, moment of inertia, tidal deformability, or oscillation spectrum.

This last point is important when interpreting phenomenological bounds. If a correction is naturally suppressed by powers of ℓ_{Pl}/L , E/E_{Pl} , or the corresponding curvature ratio, its effect in a neutron star will generally be extremely small. An observable modification therefore requires either an enhancement mechanism, a new effective scale well separated

from the Planck scale, or a coupling substantially larger than its naive Planck-suppressed value. Constraints obtained from neutron stars should accordingly be understood primarily as bounds on the effective parameters of the model being tested, rather than as direct measurements of microscopic quantum-gravity physics.

There is a second difficulty. Uncertainties in the high-density EOS can produce changes in neutron-star observables that resemble those caused by modified gravity or other new physics. A change in the radius or maximum mass, for example, is not by itself sufficient to determine whether the underlying cause lies in the matter sector or in the gravitational dynamics. Tidal observables introduce additional information, but they are also sensitive to the EOS. Reliable constraints therefore require several complementary observables and a consistent treatment of both dense-matter and gravitational uncertainties.

These considerations define the perspective adopted in the following sections. Rather than asking whether neutron stars reach the quantum gravity scale, we ask a more restricted and phenomenologically useful question: how do specific quantum-gravity-inspired corrections enter the stellar problem, what effective scales control their magnitude, and under what conditions could their consequences become observable?

3 How quantum gravity enters neutron-star calculations

Quantum-gravity-inspired physics does not translate directly into a neutron-star mass, radius, or tidal deformability. Between a microscopic modification and an astrophysical observable lies the description of dense matter, the gravitational source, and the equations governing stellar equilibrium. It is therefore useful to organize phenomenological models according to where the new physics enters the neutron-star calculation.

As summarized in Fig. 1, three conceptually distinct routes can be identified. A correction may modify the microscopic matter sector and hence the equation of state (EOS); it may alter the effective stress-energy tensor that acts as the gravitational source; or it may change the gravitational field equations themselves. These possibilities are not mutually exclusive, but separating them provides a useful framework for comparing models with otherwise very different microscopic motivations.

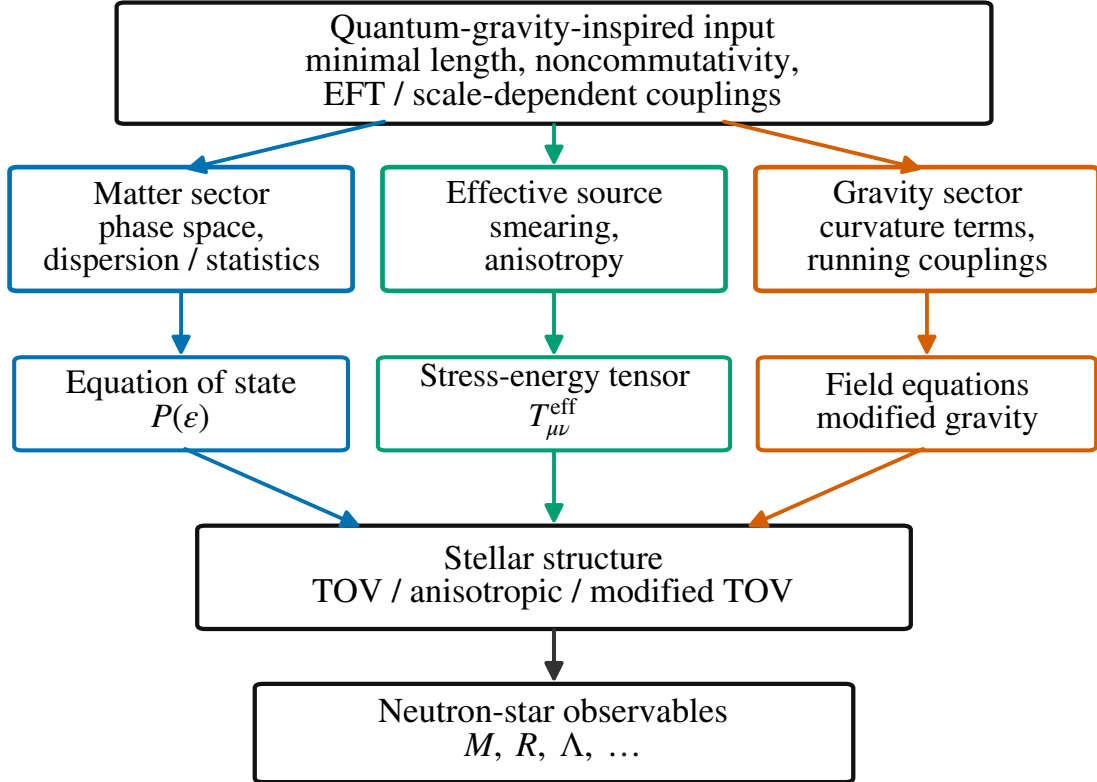
3.1 Matter-sector modifications

The first route leaves the gravitational dynamics unchanged but modifies the microscopic description of matter. Changes in the phase-space measure, dispersion relation, statistical mechanics, or interactions alter the thermodynamic quantities from which the EOS is constructed. Schematically,

$$P(\epsilon) \longrightarrow P_{\text{QG}}(\epsilon; \alpha), \quad (14)$$

where α denotes the parameter, or set of parameters, controlling the deformation.

Generalized uncertainty principles provide a representative example. Depending on the



Different microscopic mechanisms can lead to similar macroscopic signatures.

Figure 1: Schematic connection between quantum-gravity-inspired physics and neutron-star observables. New physics may enter through the microscopic matter sector, modifying the equation of state; through an effective description of the matter source, modifying the stress-energy tensor; or through the gravitational sector, modifying the field equations. Each route changes the stellar-structure problem and can ultimately affect observables such as the mass M , radius R , and tidal deformability Λ . The resulting signatures are therefore indirect and generally model dependent.

realization, a modified commutation relation changes the density of accessible momentum states and therefore the thermodynamics of degenerate matter [28, 29, 39]. Once the modified pressure and energy density have been obtained consistently, the resulting EOS can be used in the standard GR stellar-structure equations.

This route is conceptually important because it shows that a change in neutron-star structure does not necessarily imply a modification of gravity. New microscopic physics can shift the mass–radius relation or tidal response entirely through its effect on the EOS.

3.2 Effective-source modifications

A second possibility is that the new physics is represented through an effective gravitational source. In this case, the relevant modification can be written schematically as

$$T_{\mu\nu} \longrightarrow T_{\mu\nu}^{\text{eff}}(\alpha). \quad (15)$$

The gravitational field equations may retain their Einstein form, but the source entering them is no longer described by the conventional perfect-fluid stress-energy tensor.

This situation occurs naturally in phenomenological models in which a fundamental localization scale is represented by replacing a sharply localized source with an extended distribution. Depending on the construction, the effective source may also develop anisotropic stresses, so that the radial and tangential pressures are no longer equal. The stellar equilibrium equations must then be modified consistently with the form of $T_{\mu\nu}^{\text{eff}}$.

The noncommutative-inspired models discussed in Sec. 5 illustrate this route. Their role in the present article is not to provide a unique prediction of noncommutative field theory, but to show how a microscopic limitation on localization can be encoded phenomenologically in the effective source of a compact star.

3.3 Modifications of the gravitational sector

The third route changes the gravitational dynamics directly. The matter EOS may remain unchanged while the Einstein equations acquire additional terms,

$$G_{\mu\nu} + \Delta_{\mu\nu}[g; \alpha] = \frac{8\pi G}{c^4} T_{\mu\nu}, \quad (16)$$

where $\Delta_{\mu\nu}$ represents the correction generated by the effective gravitational theory. Higher-curvature operators and scale-dependent gravitational couplings are representative examples [35–38].

In this case, the stellar-structure problem itself changes. The hydrostatic-equilibrium equations must be derived from the modified field equations and their associated conservation laws rather than obtained by simply inserting a new parameter into the standard TOV equation. If tidal properties are calculated, the corresponding perturbation equations must likewise be treated within the same gravitational framework.

The three routes shown in Fig. 1 therefore modify different ingredients of the calculation,

$$\begin{aligned} \{P(\epsilon), T_{\mu\nu}^{\text{eff}}, \text{field equations}\} &\longrightarrow \text{stellar structure} \\ &\longrightarrow \{M, R, \Lambda, \dots\}. \end{aligned} \quad (17)$$

This separation will be used throughout the remainder of the article. It also makes clear why astrophysical constraints are inherently model dependent: physically different microscopic mechanisms can produce similar changes in the same macroscopic observables.

3.4 The equation of state as the microscopic input

For corrections that enter through the matter sector, the EOS provides the bridge between microscopic physics and stellar structure [4–7]. Although observations constrain global quantities such as masses and radii, these quantities depend on the pressure generated by matter over a broad range of densities. Any microscopic deformation that changes the

Table 1: Three entry points for quantum-gravity-inspired corrections.

Route	Modified object	Control parameter	Minimum consistency requirement
GUP matter sector	phase space, dispersion, or statistics	βp_F^2	derive n , ε , and P from one potential and solve the interacting composition self-consistently
NC-inspired source	$T_{\mu\nu}^{\text{eff}}$ and localization	θ/L^2	define the kernel, close the pressure sector, and impose $\nabla_\mu T_{\text{eff}}^{\mu\nu} = 0$
Gravitational EFT	field equations and perturbations	$\mathcal{R}/M^2$	control the expansion, remove spurious branches, and use the same theory for the background and tides

pressure, energy density, or composition can therefore propagate to observable neutron-star properties.

For matter in thermodynamic equilibrium, it is convenient to introduce the grand-potential density

$$\omega(T, \mu) = \frac{\Omega}{V}, \quad (18)$$

from which

$$P = -\omega, \quad n = -\left(\frac{\partial \omega}{\partial \mu}\right)_T, \quad (19)$$

and

$$\epsilon = \omega + Ts + \mu n. \quad (20)$$

Here n and s are the particle-number and entropy densities, respectively. In the zero-temperature limit relevant for cold neutron-star matter,

$$\epsilon = \omega + \mu n. \quad (21)$$

Eliminating the microscopic variables then yields the relation $P = P(\epsilon)$ required by the stellar-structure equations.

A relativistic degenerate Fermi gas provides a transparent illustration. At zero temperature,

$$\omega = -g \int \frac{d^3 p}{(2\pi\hbar)^3} [\mu - E(p)] \Theta(p_F - p), \quad (22)$$

with

$$E(p) = \sqrt{p^2 c^2 + m^2 c^4}, \quad (23)$$

where g is the degeneracy factor and p_F is the Fermi momentum. For the ideal gas at $T = 0$, $\mu = E(p_F)$.

Equation (22) makes explicit the microscopic ingredients that can be modified. A deformation of the phase-space measure changes the weight assigned to momentum states, a modified dispersion relation changes their single-particle energies, and new interactions

contribute directly to the thermodynamic potential. These changes need not lead to the same EOS even when they arise from related ideas at the microscopic level.

Realistic neutron-star matter is considerably more complicated than an ideal one-component Fermi gas. Strong interactions, the particle composition, charge neutrality, and beta equilibrium all contribute to the EOS. Equation (22) should therefore be viewed only as a simple baseline for identifying where a microscopic deformation enters. In a realistic calculation, the same deformation must be incorporated consistently into all thermodynamic quantities that it affects.

The remainder of the article follows the three routes introduced above. Generalized uncertainty principles illustrate modifications of the matter sector; noncommutative-inspired constructions illustrate changes in the effective source; and the effective-field-theory treatment of gravity illustrates modifications of the gravitational dynamics. This organization makes it possible to compare their neutron-star consequences within a common framework while keeping their physical origins distinct.

4 Generalized uncertainty principle

A recurring idea in quantum-gravity phenomenology is that the notion of arbitrarily short distances may cease to be physically meaningful. Arguments based on quantum gravity, string-inspired models, and thought experiments combining quantum mechanics with gravitational collapse have motivated the possibility of an effective minimum measurable length [1, 3, 28, 29]. Generalized uncertainty principles (GUPs) provide a convenient phenomenological framework for describing such a possibility. They should not, in general, be regarded as a complete microscopic theory of quantum spacetime, but rather as effective deformations through which possible short-distance effects can be studied at lower energies.

In ordinary quantum mechanics, the canonical algebra

$$[x_i, p_j] = i\hbar\delta_{ij} \quad (24)$$

leads to the Heisenberg uncertainty relation

$$\Delta x \Delta p \geq \frac{\hbar}{2}. \quad (25)$$

Within this framework, the uncertainty in position can in principle be made arbitrarily small by increasing the uncertainty in momentum. Gravity changes the physical interpretation of this limiting procedure: localizing a state within an increasingly small region requires an increasing concentration of energy, which must eventually contribute non-negligibly to the surrounding spacetime geometry [1, 40]. This observation motivates deformations of the canonical algebra in which momentum-dependent corrections become important at sufficiently high momentum.

A commonly used schematic realization is

$$[x_i, p_j] = i\hbar\delta_{ij} \left(1 + \beta p^2 + \dots\right), \quad (26)$$

where β controls the strength of the deformation. This expression is deliberately simplified. In more general multidimensional constructions, the commutator can contain additional tensor structures, including terms proportional to $p_i p_j$, and may involve more than one deformation parameter [3, 29]. Consequently, neither the precise form of the algebra nor its phenomenological consequences are universal among GUP models.

At the level of the uncertainty relation, Eq. (26) suggests a modification of the form

$$\Delta x \Delta p \gtrsim \frac{\hbar}{2} \left[1 + \beta(\Delta p)^2 + \dots\right]. \quad (27)$$

Minimizing the right-hand side with respect to the momentum uncertainty gives a characteristic minimum position uncertainty of order

$$(\Delta x)_{\min} \sim \hbar\sqrt{\beta}. \quad (28)$$

It is customary to parametrize the deformation as

$$\beta = \frac{\beta_0}{(M_{\text{Pl}}c)^2}, \quad (29)$$

where β_0 is dimensionless and M_{Pl} is the unreduced Planck mass. The parameter β therefore has dimensions of inverse momentum squared, and the relevant expansion parameter is βp^2 .

If β_0 is of order unity, the deformation becomes appreciable only for momenta approaching the Planck scale. For characteristic neutron-star momenta, this produces an enormous suppression. Taking $p \sim 1 \text{ GeV}/c$ only as an order-of-magnitude scale,

$$\beta p^2 = \beta_0 \left(\frac{pc}{M_{\text{Pl}}c^2}\right)^2 \sim 10^{-38} \beta_0. \quad (30)$$

Thus, for $\beta_0 \sim \mathcal{O}(1)$, a Planck-suppressed GUP correction is effectively invisible in neutron-star matter. Observable stellar effects require either a very large effective value of β_0 or a deformation associated with a scale substantially below the Planck scale. Astrophysical constraints on β should therefore be interpreted primarily as bounds on an effective GUP realization, rather than as direct measurements of the Planck length [3, 44]. An idealized GUP Fermi-gas analysis obtained $\beta_0 \lesssim 10^{37}$ from neutron-star masses, while emphasizing that nuclear interactions can produce comparable macroscopic changes [44]. This limit is conditional on the adopted matter model; it is not evidence that a parameter of natural size has become observable.

4.1 Modified phase-space measure

One possible consequence of a deformed quantum algebra is a modification of the density of available states in momentum space. This is particularly relevant for degenerate matter because its thermodynamic properties are determined by integrals over occupied momentum states. A deformation of the phase-space measure can therefore modify the equation of state even when the single-particle dispersion relation and the gravitational field equations are left unchanged.

The relation between a GUP algebra and the corresponding phase-space measure depends on the representation chosen for the deformed commutation relations. There is no unique measure associated with all GUP models. To make the thermodynamic consequences explicit, we adopt the representative isotropic prescription

$$\frac{d^3p}{(2\pi\hbar)^3} \longrightarrow \frac{d^3p}{(2\pi\hbar)^3(1 + \beta p^2)^3}. \quad (31)$$

The additional factor suppresses the contribution of high-momentum states. Equation (31) should therefore be understood as a particular phenomenological realization of GUP-modified phase space, rather than as a model-independent prediction of the generalized uncertainty principle [3, 29]. For this algebra, the cubic power follows from the invariant three-dimensional phase-space volume obtained from the corresponding classical Liouville theorem [45]; changing the algebra or the number of dimensions generally changes the Jacobian.

To display the deformation independently of a particular momentum unit, we introduce a reference scale p_0 and define

$$x \equiv \frac{p}{p_0}, \quad \tilde{\beta} \equiv \beta p_0^2. \quad (32)$$

The phase-space weight then becomes

$$w(x) = (1 + \tilde{\beta} x^2)^{-3}. \quad (33)$$

Figure 2 shows this weight for several representative values of $\tilde{\beta}$. The undeformed limit is recovered for $\tilde{\beta} = 0$, whereas increasing the deformation progressively suppresses states at large momentum.

The thermodynamic consequences can be seen most transparently for a relativistic Fermi gas at zero temperature. In the undeformed case, the number density is

$$n = \frac{g}{2\pi^2\hbar^3} \int_0^{p_F} p^2 dp = \frac{gp_F^3}{6\pi^2\hbar^3}, \quad (34)$$

where g is the degeneracy factor and p_F is the Fermi momentum. The corresponding

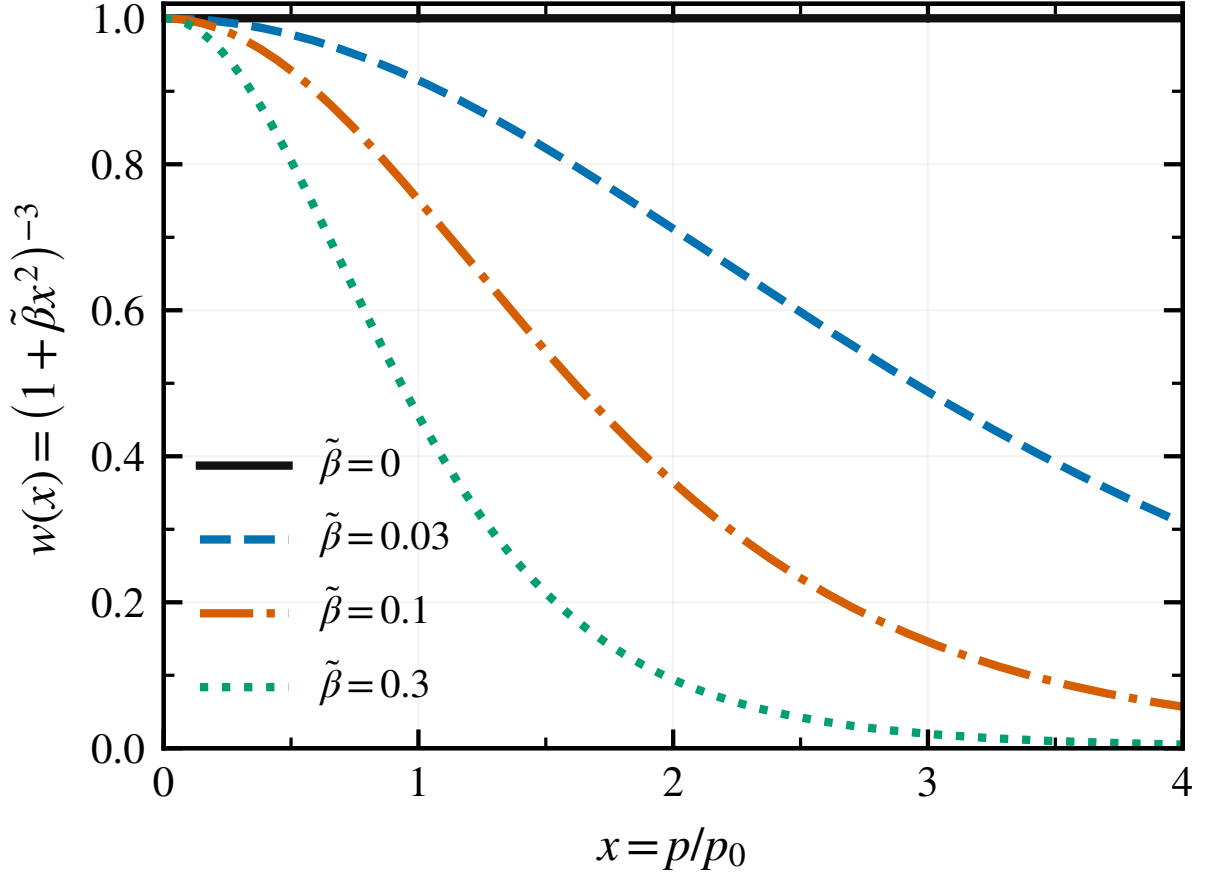


Figure 2: Dimensionless phase-space weight $w(x) = (1 + \tilde{\beta}x^2)^{-3}$ for representative values of the deformation parameter $\tilde{\beta} = \beta p_0^2$, with $x = p/p_0$. The canonical phase-space measure is recovered for $\tilde{\beta} = 0$. Positive values of $\tilde{\beta}$ progressively suppress high-momentum states. The curves illustrate the specific phase-space realization adopted in Eq. (31) and should not be interpreted as a universal prediction of GUP models.

energy density and pressure are

$$\epsilon = \frac{g}{2\pi^2\hbar^3} \int_0^{p_F} p^2 E(p) dp, \quad (35)$$

$$P = \frac{g}{6\pi^2\hbar^3} \int_0^{p_F} \frac{p^4 c^2}{E(p)} dp, \quad (36)$$

with

$$E(p) = \sqrt{p^2 c^2 + m^2 c^4}. \quad (37)$$

Using the modified measure in Eq. (31), the particle-number density becomes

$$n_{\text{GUP}} = \frac{g}{2\pi^2\hbar^3} \int_0^{p_F} \frac{p^2 dp}{(1 + \beta p^2)^3}, \quad (38)$$

while the energy density is

$$\epsilon_{\text{GUP}} = \frac{g}{2\pi^2\hbar^3} \int_0^{p_F} \frac{p^2 E(p)}{(1 + \beta p^2)^3} dp. \quad (39)$$

The pressure is most conveniently obtained from the same thermodynamic potential rather than introduced through an independent deformation of the kinetic-pressure integral. At zero temperature, the grand-potential density for the noninteracting gas is

$$\omega_{\text{GUP}} = -\frac{g}{2\pi^2\hbar^3} \int_0^{p_F} \frac{p^2[\mu - E(p)]}{(1 + \beta p^2)^3} dp, \quad (40)$$

where

$$\mu = E(p_F). \quad (41)$$

The pressure follows from

$$P_{\text{GUP}} = -\omega_{\text{GUP}}, \quad (42)$$

and the number density is obtained consistently from

$$n_{\text{GUP}} = -\left(\frac{\partial\omega_{\text{GUP}}}{\partial\mu}\right)_{T=0}. \quad (43)$$

These expressions recover the ordinary relativistic Fermi-gas thermodynamics in the limit $\beta \rightarrow 0$.

The deformation changes the relation among n , ϵ , and P and therefore modifies the EOS. In the class of models considered here, this is the point at which the GUP enters the neutron-star calculation: the microscopic thermodynamics is modified, while the Einstein field equations and the standard TOV equations remain unchanged.

4.2 Perturbative limit and characteristic size of the correction

The magnitude of the effect is particularly transparent in the perturbative regime

$$\beta p_F^2 \ll 1. \quad (44)$$

Expanding the phase-space weight gives

$$(1 + \beta p^2)^{-3} = 1 - 3\beta p^2 + 6\beta^2 p^4 + \mathcal{O}(\beta^3 p^6). \quad (45)$$

Keeping the leading correction in Eq. (38) yields

$$n_{\text{GUP}} \simeq \frac{g}{2\pi^2\hbar^3} \left(\frac{p_F^3}{3} - \frac{3\beta p_F^5}{5} \right). \quad (46)$$

Relative to the undeformed density evaluated at the same Fermi momentum,

$$n = \frac{gp_F^3}{6\pi^2\hbar^3}, \quad (47)$$

the leading fractional correction is

$$\frac{\delta n}{n} \equiv \frac{n_{\text{GUP}} - n}{n} \simeq -\frac{9}{5}\beta p_F^2. \quad (48)$$

The sign reflects the suppression of the available phase-space volume in the realization adopted here, while the magnitude is controlled by the dimensionless combination βp_F^2 .

Equation (48) refers specifically to a comparison performed at fixed p_F . This distinction matters when constructing an EOS. If the physical baryon density n_B is specified instead, the Fermi momentum is no longer the same in the deformed and undeformed systems. It must be determined self-consistently from

$$n_B = \frac{g}{2\pi^2\hbar^3} \int_0^{p_F} \frac{p^2 dp}{(1 + \beta p^2)^3}. \quad (49)$$

The corresponding ϵ_{GUP} and P_{GUP} must then be evaluated using this value of p_F . A fixed- p_F expansion is useful for identifying the scale and sign of the microscopic correction, but it should not be confused with a self-consistent comparison of equations of state at fixed physical density.

4.3 Connection with realistic dense-matter models

The free Fermi-gas calculation isolates the effect of the modified phase-space measure, but it is not a realistic model of a neutron-star core. Dense stellar matter involves strong interactions, several particle species, beta equilibrium, charge neutrality, and many-body correlations, all of which contribute to the EOS [5–7].

A realistic implementation must therefore introduce the deformation consistently within the chosen microscopic model. Schematically, the grand-potential density may take the form

$$\omega_{\text{GUP}} = -g \int \frac{d^3p}{(2\pi\hbar)^3(1 + \beta p^2)^3} \mathcal{F}(p, \mu, M^*, \dots) + \omega_{\text{int}}, \quad (50)$$

where $\mathcal{F}$ represents the appropriate single-particle contribution and ω_{int} contains the interaction terms. Equation (50) is schematic by construction. Depending on the microscopic theory and on the chosen GUP realization, the deformation may also enter effective masses, mean fields, particle fractions, chemical-equilibrium conditions, and the self-consistency equations that determine the state of the matter.

This dependence on the underlying dense-matter model is important when astrophysical bounds are interpreted. A GUP-induced change in the EOS can be partially degenerate with uncertainties in nuclear interactions, many-body correlations, the particle content of the core, or possible high-density phase transitions. A bound on β obtained from a mass, radius, or tidal-deformability measurement is therefore also conditional on the microscopic EOS used in the calculation.

For this reason, neutron-star constraints on GUP-inspired physics are most meaningful when three requirements are satisfied simultaneously: the deformation is implemented

thermodynamically consistently, the underlying dense-matter model is realistic, and several complementary astrophysical observables are considered. Within such a framework, neutron stars can constrain the effective scale associated with the deformation, even though the natural Planck-suppressed correction itself is far too small to produce an observable stellar effect [5–7, 23, 24].

5 Noncommutative geometry and smeared matter distributions

Noncommutative geometry provides a conceptually different route from the generalized uncertainty principle discussed in the previous section. Rather than modifying the phase-space structure of matter, the basic idea is that spacetime coordinates themselves need not commute at sufficiently short distances. A commonly used schematic relation is

$$[\hat{x}^\mu, \hat{x}^\nu] = i\theta^{\mu\nu}, \quad (51)$$

where $\theta^{\mu\nu}$ is an antisymmetric tensor characterizing the noncommutative structure [3, 30]. Its components have dimensions of length squared and therefore introduce characteristic noncommutative scales of order $\sqrt{|\theta^{\mu\nu}|}$.

Equation (51) should be understood as a spacetime relation. A complete noncommutative field theory generally requires a deformation of the product of fields, for example through a star product, and its phenomenology depends on the particular realization [30]. In applications to static compact objects, however, one often adopts a simpler effective description in which the loss of perfect localization is represented by a spatial smearing of the matter source. The discussion below refers to this noncommutative-inspired construction and should not be interpreted as a complete four-dimensional noncommutative theory of neutron-star matter.

The simplest example is a pointlike source. In a frequently used prescription, the three-dimensional Dirac distribution is replaced by a Gaussian kernel,

$$\delta^{(3)}(\mathbf{r}) \longrightarrow K_\theta(r) = \frac{1}{(4\pi\theta)^{3/2}} \exp\left(-\frac{r^2}{4\theta}\right), \quad (52)$$

where $\theta > 0$ sets the effective smearing scale. A point mass M is then represented by

$$\rho_\theta(r) = MK_\theta(r) = \frac{M}{(4\pi\theta)^{3/2}} \exp\left(-\frac{r^2}{4\theta}\right). \quad (53)$$

The characteristic width of the source is therefore of order $\sqrt{\theta}$. In the formal limit

$$\theta \rightarrow 0, \quad (54)$$

the Gaussian kernel approaches the ordinary Dirac distribution and the point-source

description is recovered.

For a neutron star, however, the situation is more subtle. The stellar interior is already described by an extended energy-density profile $\epsilon(r)$ rather than by a point source. Consequently, Eq. (53) cannot by itself be taken as a unique noncommutative prediction for the matter distribution of a star.

A natural phenomenological extension is to smear the original energy density according to

$$\epsilon_\theta(\mathbf{r}) = \int d^3r' K_\theta(|\mathbf{r} - \mathbf{r}'|) \epsilon(\mathbf{r}'), \quad (55)$$

where K_θ is the Gaussian kernel introduced in Eq. (52). This form makes the nonlocal character of the prescription explicit: the effective density at a given point receives contributions from a finite surrounding region whose size is controlled by $\sqrt{\theta}$.

Equation (55) is useful as a coordinate-space model, but it is not by itself a covariant prescription for a relativistic star. In a fully geometric treatment, both the integration measure and the distance entering the kernel should be defined with respect to the spatial geometry of the stellar configuration. The expression above should therefore be regarded as a transparent phenomenological realization of nonlocal smearing rather than as a unique covariant consequence of Eq. (51).

For a static and spherically symmetric configuration, one may define an effective enclosed mass from the smeared density,

$$m_\theta(r) = 4\pi \int_0^r \frac{\epsilon_\theta(r')}{c^2} r'^2 dr'. \quad (56)$$

This illustrates directly how a redistribution of the energy density can modify the mass profile that enters the stellar structure equations. However, the mass function alone is not sufficient to define a self-consistent equilibrium configuration.

The full effective stress-energy tensor must satisfy the conservation condition

$$\nabla_\mu T_{\text{eff}}^{\mu\nu} = 0. \quad (57)$$

Moreover, the effective source generated by a noncommutative-inspired construction need not retain the perfect-fluid form. In particular, the radial and tangential pressures may differ,

$$P_r(r) \neq P_t(r), \quad (58)$$

so that the stress-energy tensor becomes anisotropic.

This has an immediate consequence for the stellar equilibrium problem. For an anisotropic fluid, the radial conservation equation contains an additional contribution proportional to $P_t - P_r$, and the usual perfect-fluid TOV equation must be replaced by its anisotropic form. Therefore, smearing the density profile while leaving the pressure sector unchanged is not sufficient. The energy density, pressure components, and conservation equations must be constructed consistently within the same effective model.

This point is particularly important because the Gaussian smearing prescription is not

unique. Different noncommutative-inspired constructions can generate different effective density profiles, pressure anisotropies, and therefore different stellar structure equations. The parameter θ should consequently be understood as a model-dependent effective scale rather than as a universal noncommutative parameter whose astrophysical interpretation is independent of the underlying construction.

The scale hierarchy again determines whether the effect can become observable. If the noncommutative length is genuinely of order the Planck length,

$$\sqrt{\theta} \sim \ell_{\text{Pl}}, \quad (59)$$

the corresponding modification is expected to be negligible on neutron-star length scales. Observable changes in the stellar structure would instead require an effective smearing scale much larger than ℓ_{Pl} , or some additional mechanism capable of enhancing the low-energy manifestation of the noncommutative physics. Bounds derived from neutron stars should therefore be interpreted as constraints on an effective noncommutative-inspired model rather than as direct measurements of microscopic spacetime noncommutativity [3, 30].

It is also useful to separate the smearing prescription from the microscopic equation of state. The local thermodynamic relation

$$P = P(\epsilon) \quad (60)$$

is determined by the underlying description of dense matter, including nuclear interactions, particle composition, beta equilibrium, charge neutrality, and many-body effects [5–7]. In the class of models discussed here, noncommutative-inspired effects act instead through the effective spatial distribution and stress-energy structure of the source.

A realistic neutron-star calculation must therefore combine two ingredients consistently: a microscopic EOS for dense matter and an effective stress-energy tensor implementing the chosen noncommutative-inspired prescription. Only after both are specified can one solve the corresponding stellar structure equations and determine observable quantities such as the mass, radius, or tidal response.

The comparison with the GUP case highlights the usefulness of the classification introduced in Sec. 3. GUP-inspired models can modify the microscopic thermodynamics and hence the EOS, whereas the noncommutative-inspired construction considered here acts primarily through the localization and stress-energy structure of the source. The next section considers the third route, in which the gravitational dynamics themselves are modified through higher-curvature terms and other effective extensions of GR.

6 Effective field theory and modified gravity

The previous sections considered corrections that enter through the microscopic matter sector or through the effective stress-energy source. A third possibility is that the gravitational dynamics themselves are modified. At energies well below the scale at which the underlying

high-energy theory becomes relevant, such effects can be organized systematically within an effective field theory (EFT) description of gravity [35, 36].

The EFT approach does not require a complete knowledge of the ultraviolet theory. Instead, one writes the most general low-energy action compatible with the symmetries of the theory and organizes the allowed operators according to their number of derivatives, or equivalently their mass dimension. For gravity, general covariance allows scalar combinations of curvature tensors and their covariant derivatives. The Einstein–Hilbert term provides the leading contribution at low curvature, while higher-curvature and higher-derivative operators encode corrections associated with shorter distance scales.

Throughout this section we use natural units,

$$\hbar = c = 1. \quad (61)$$

A convenient schematic form of the local gravitational EFT is

$$S_{\text{EFT}} = \frac{\bar{M}_{\text{Pl}}^2}{2} \int d^4x \sqrt{-g} \left[R + \frac{c_1}{M^2} R^2 + \frac{c_2}{M^2} R_{\mu\nu} R^{\mu\nu} + \frac{d_1}{M^4} \mathcal{O}_6 + \frac{d_2}{M^6} \mathcal{O}_8 + \dots \right] + S_{\text{matter}}, \quad (62)$$

where

$$\bar{M}_{\text{Pl}} = \frac{1}{\sqrt{8\pi G}} \quad (63)$$

is the reduced Planck mass. The scale M characterizes the heavy degrees of freedom that are not resolved explicitly in the low-energy description, while c_i and d_i are dimensionless Wilson coefficients. The quantities $\mathcal{O}_6$ and $\mathcal{O}_8$ represent local curvature invariants of mass dimension six and eight, respectively. For example, $\mathcal{O}_6$ may contain terms cubic in the curvature.

The dimensional hierarchy makes the derivative expansion transparent. In natural units, the Ricci scalar has mass dimension two,

$$[R] = (\text{mass})^2, \quad (64)$$

and therefore

$$[R^2] = (\text{mass})^4, \quad [\mathcal{O}_6] = (\text{mass})^6, \quad [\mathcal{O}_8] = (\text{mass})^8. \quad (65)$$

The inverse powers of the heavy scale M therefore suppress operators containing additional powers of the curvature or additional derivatives.

Equation (62) is useful for power counting, but it should not be interpreted as a unique or minimal operator basis. In four spacetime dimensions, the Gauss–Bonnet identity relates the quadratic curvature invariants, while field redefinitions can move contributions between different operators. In the presence of matter, such redefinitions may also relate

curvature terms to interactions involving the matter fields [35, 36].

Varying the action with respect to the metric leads schematically to

$$G_{\mu\nu} + \Delta_{\mu\nu} = \frac{1}{\bar{M}_{\text{Pl}}^2} T_{\mu\nu} = 8\pi G T_{\mu\nu}, \quad (66)$$

where $\Delta_{\mu\nu}$ collects the contributions generated by the higher-order operators. Its explicit form depends on the terms retained in the effective action. In this class of models, the microscopic equation of state may remain unchanged even though the relation between the matter source and the spacetime geometry is modified.

A familiar example is

$$f(R) = R + \alpha R^2, \quad (67)$$

where α has dimensions of inverse mass squared, or equivalently length squared. This expression provides a simple illustration of how a curvature-squared contribution can modify the gravitational field equations while preserving general covariance [37, 38].

An important distinction must be made between using Eq. (67) as part of a low-energy expansion and treating it as a complete modified-gravity theory. In the EFT interpretation, the R^2 term is a correction whose importance is controlled by the ratio between the characteristic curvature and the scale suppressing the operator. The same functional form can instead be studied nonperturbatively as a particular theory of gravity. Here it is used only to illustrate how corrections in the gravitational sector can enter the neutron-star problem.

More general theories may contain additional curvature invariants, covariant derivatives, extra propagating degrees of freedom, or nonminimal couplings to matter. There is therefore no universal “modified TOV equation.” The appropriate stellar-structure equations must be derived from the field equations and conservation laws of the specific theory under consideration.

At the level of hydrostatic equilibrium, a departure from GR may be represented schematically as

$$\frac{dP}{dr} = F_{\text{GR}} + \delta F, \quad (68)$$

where F_{GR} denotes the standard GR hydrostatic-balance term and δF represents the correction generated by the modified field equations. The quantity δF is not an arbitrary additional force: it must follow consistently from the underlying theory and its associated conservation equations.

Changes in the gravitational dynamics can shift the mass–radius relation, modify the maximum supported mass, and alter the compactness and tidal response of the star [33, 34]. Figure 3 illustrates two generic consequences that are particularly relevant for observations: a displacement of the radius at fixed mass and a change in the maximum mass.

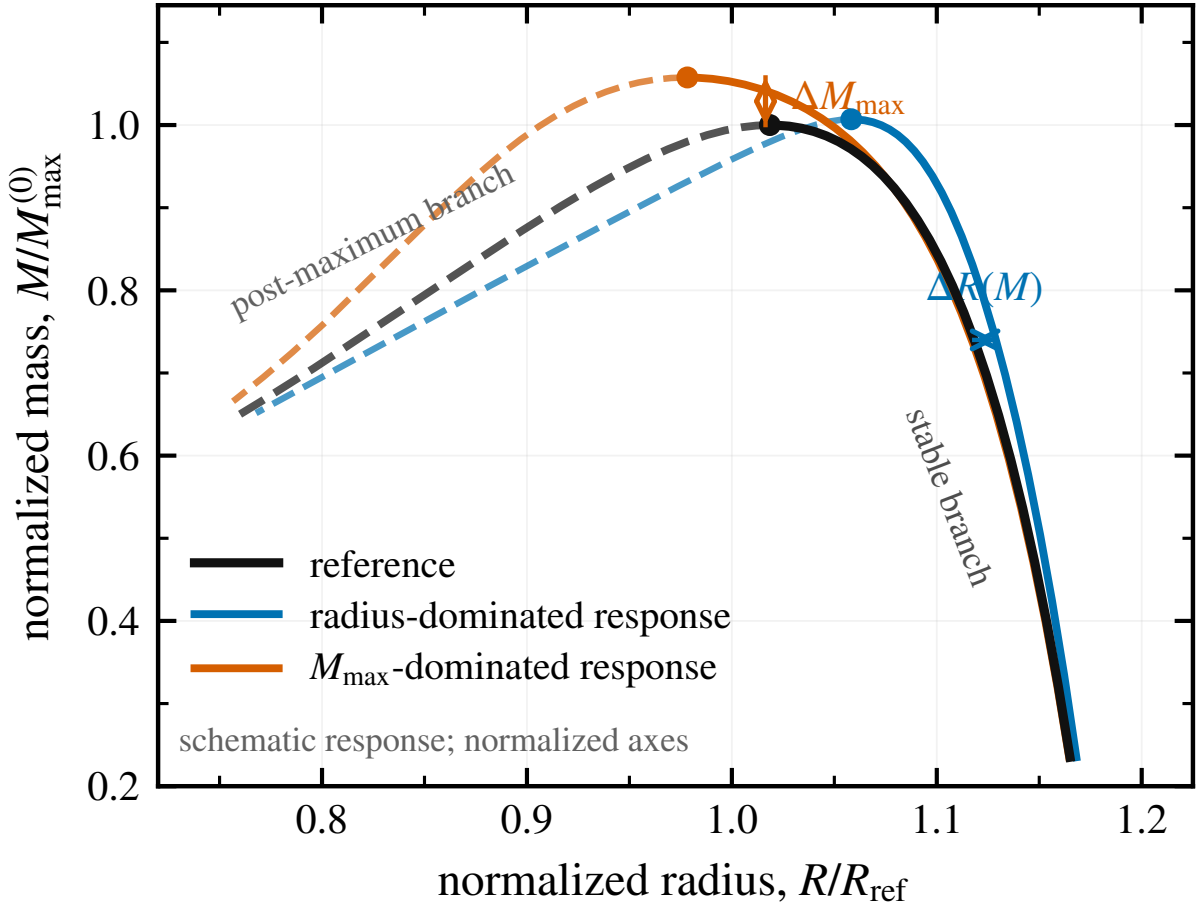


Figure 3: Schematic response of a neutron-star mass–radius sequence to changes in the underlying stellar model. The black curve represents a normalized reference sequence, while the additional curves illustrate a predominantly radius-dominated response and a modification of the maximum supported mass. Solid portions denote the stable branch and dashed portions the post-maximum branch. The axes are normalized and the curves are illustrative only; they are not solutions obtained from a specific equation of state or quantum-gravity-inspired model.

6.1 Power counting and the expected size of the corrections

A central advantage of the EFT framework is that it provides a direct estimate of the expected size of higher-order corrections. To distinguish the characteristic curvature scale from the Ricci scalar R and from the stellar radius, we denote it by $\mathcal{R}$. In natural units,

$$[\mathcal{R}] = (\text{mass})^2. \quad (69)$$

The Einstein–Hilbert contribution therefore scales as

$$\mathcal{L}_{\text{EH}} \sim \bar{M}_{\text{Pl}}^2 \mathcal{R}. \quad (70)$$

A curvature-squared term in Eq. (62) scales schematically as

$$\delta\mathcal{L}_{R^2} \sim \bar{M}_{\text{Pl}}^2 \frac{c_i}{M^2} \mathcal{R}^2, \quad (71)$$

so that, relative to the Einstein–Hilbert term,

$$\frac{\delta\mathcal{L}_{R^2}}{\mathcal{L}_{\text{EH}}} \sim c_i \frac{\mathcal{R}}{M^2}. \quad (72)$$

The dimensionless quantity

$$\frac{\mathcal{R}}{M^2} \quad (73)$$

therefore controls the derivative expansion. When $\mathcal{R} \ll M^2$, curvature-squared corrections remain perturbative unless the associated Wilson coefficient is parametrically large.

The same reasoning applies to higher powers of the curvature. A local operator containing n powers of the curvature scales schematically as

$$\delta\mathcal{L}_{R^n} \sim \bar{M}_{\text{Pl}}^2 \frac{c_n}{M^{2n-2}} \mathcal{R}^n, \quad n \geq 2, \quad (74)$$

which gives

$$\frac{\delta\mathcal{L}_{R^n}}{\mathcal{L}_{\text{EH}}} \sim c_n \left(\frac{\mathcal{R}}{M^2} \right)^{n-1}. \quad (75)$$

For instance, a curvature-cubed contribution is suppressed by two powers of the same expansion parameter,

$$\frac{\delta\mathcal{L}_{R^3}}{\mathcal{L}_{\text{EH}}} \sim c_3 \left(\frac{\mathcal{R}}{M^2} \right)^2. \quad (76)$$

Higher-curvature terms are therefore progressively suppressed as long as the characteristic curvature remains well below the scale M^2 .

If the scale associated with new physics is of order the reduced Planck mass,

$$M \sim \bar{M}_{\text{Pl}}, \quad (77)$$

the leading curvature-squared correction becomes

$$\frac{\delta\mathcal{L}_{R^2}}{\mathcal{L}_{\text{EH}}} \sim c_i \frac{\mathcal{R}}{\bar{M}_{\text{Pl}}^2}. \quad (78)$$

For neutron-star spacetimes,

$$\frac{\mathcal{R}}{\bar{M}_{\text{Pl}}^2} \ll 1, \quad (79)$$

so corrections suppressed directly by the Planck scale are expected to be extremely small when the Wilson coefficients are of natural size.

This is the gravitational analogue of the scale hierarchy encountered in the GUP and noncommutative-inspired cases. Neutron stars are strongly relativistic on astrophysical scales, but their characteristic curvatures remain far below the Planck scale. Observable

deviations therefore require an additional ingredient, such as a lower effective scale M , enhanced Wilson coefficients, additional light degrees of freedom, or another mechanism that amplifies the low-energy response.

Truncating a higher-derivative action and then solving it nonperturbatively can introduce solution branches outside the EFT. A controlled treatment retains solutions continuously connected to GR, commonly through order reduction or an equivalent expansion. This prevents degrees of freedom created by the truncation from being mistaken for low-energy predictions [46].

The local curvature expansion is not the complete gravitational EFT. Operators containing covariant derivatives of the curvature appear at higher orders, while quantum loops generate nonlocal contributions that cannot be represented by a finite polynomial of local curvature invariants [35, 36]. The discussion above is therefore intended only to describe the power counting of the local derivative expansion relevant to the present analysis.

6.2 Scale-dependent gravitational couplings

A conceptually different possibility arises in renormalization-group approaches to gravity, where Newton’s coupling depends on a renormalization scale,

$$G \longrightarrow G(k), \tag{80}$$

with k denoting the RG scale [31, 32]. This should be distinguished from the local EFT expansion in Eq. (62). The latter organizes higher-derivative operators in the effective action, whereas $G(k)$ describes the scale dependence of a coupling under renormalization-group evolution.

Applying a running gravitational coupling to a neutron star requires an additional physical prescription because the RG scale k must be related to a characteristic scale of the stellar configuration. Possible choices include a local curvature invariant, a density scale, or an inverse characteristic length. There is no unique identification, and different scale-setting prescriptions can lead to different stellar solutions.

For the same reason, it is generally not sufficient to replace G by a position-dependent function directly in the standard Einstein equations. The resulting field equations must remain compatible with the Bianchi identities and with the appropriate conservation equations for matter and the effective gravitational sector. A consistent stellar calculation therefore requires three ingredients: the running law $G(k)$, a prescription relating k to the stellar configuration, and the corresponding modified field equations.

Once these ingredients are specified consistently, the resulting equilibrium solutions can be confronted with neutron-star observables such as masses, radii, compactnesses, and tidal deformabilities [33, 34]. As in the previous examples, the resulting bounds apply to the particular low-energy realization being tested and should not automatically be interpreted as direct measurements of Planck-scale quantum gravity.

7 Observational channels and prospective constraints

The phenomenological relevance of the quantum-gravity-inspired scenarios discussed in the previous sections ultimately depends on whether their effects can be distinguished from uncertainties in conventional neutron-star physics. This requires connecting the modified microscopic or gravitational description to quantities that can be measured astrophysically. The most useful information currently comes from three complementary channels: precision measurements of neutron-star masses, X-ray constraints on masses and radii, and gravitational-wave measurements of the tidal response of neutron stars in compact binaries.

For a specified microscopic EOS, effective source, and theory of gravity, the stellar structure equations determine a family of equilibrium configurations. From this sequence one obtains quantities such as the mass M , radius R , compactness C , maximum supported mass M_{max} , and tidal deformability Λ . A quantum-gravity-inspired parameter therefore enters the comparison with observations only indirectly. Schematically,

$$\alpha_{\text{QG}} \longrightarrow \{P(\epsilon), T_{\mu\nu}^{\text{eff}}, \text{gravity}\} \longrightarrow \{M, R, \Lambda, \dots\}. \quad (81)$$

If the deformation acts only on the matter sector, the modified EOS is inserted into the standard GR stellar equations. If the gravitational sector is changed, both the equilibrium equations and, where relevant, their perturbations must be derived and solved within the same theory. Likewise, an effective modification of the matter source must be implemented consistently in the stress-energy tensor entering the stellar problem.

7.1 Mass constraints

Neutron-star masses provide one of the most direct constraints on stellar models because several pulsar masses are known with high precision. The existence of neutron stars with masses close to or above $2 M_{\odot}$ is now firmly established [10–13]. Representative examples include

$$M_{\text{J1614-2230}} \simeq 1.97 M_{\odot}, \quad (82)$$

$$M_{\text{J0348+0432}} \simeq 2.01 M_{\odot}, \quad (83)$$

$$M_{\text{J0740+6620}} \simeq 2.08 M_{\odot}. \quad (84)$$

These measurements imply the robust requirement

$$M_{\text{max}} \gtrsim 2 M_{\odot} \quad (85)$$

for any viable combination of dense-matter physics and gravitational dynamics.

Additional information comes from very massive spider pulsars. PSR J0952–0607 has been reported with a mass around $2.35 M_{\odot}$, although its inference is more model dependent than the Shapiro-delay measurements of pulsar–white-dwarf systems [14]. Such

systems provide potentially stronger constraints on the upper end of the neutron-star mass distribution, but it is useful to distinguish them from the more robust $\sim 2 M_\odot$ lower bound.

A quantum-gravity-inspired correction that strongly softens the effective EOS or increases the gravitational attraction sufficiently to lower M_{max} below the observed range is therefore disfavored. The interpretation is nevertheless not unique. A modification of gravity that lowers M_{max} can, in principle, be compensated by a sufficiently stiff microscopic EOS, while a change in the matter sector can mimic the effect of altered gravitational dynamics. Mass measurements are therefore powerful consistency tests, but they cannot by themselves identify the microscopic origin of a deviation.

7.2 Radius measurements

Radius measurements provide complementary information because they probe the pressure of dense matter over a different range of densities from that controlling the maximum mass. X-ray pulse-profile modeling with the Neutron Star Interior Composition Explorer (NICER) has provided simultaneous mass–radius constraints for several millisecond pulsars. The first major results were obtained for PSR J0030+0451 and the massive pulsar PSR J0740+6620 [15–18].

The observational picture has continued to develop. Updated analyses of PSR J0740+6620 using substantially larger NICER data sets have refined its radius constraint [20, 21], while NICER observations of the nearby pulsar PSR J0437–4715 have provided an independent radius measurement at a mass close to $1.4 M_\odot$ [19]. These newer results broaden the mass range over which the neutron-star mass–radius relation can be tested directly.

Taken together with nuclear and gravitational-wave information, current analyses typically favor radii near ~ 12 km for canonical-mass neutron stars, with uncertainties at the kilometer level and some dependence on the EOS parametrization and adopted data sets [22, 42, 43]. It is therefore more appropriate to retain a broad characteristic scale,

$$R_{\text{NS}} \sim 11\text{--}14 \text{ km}, \tag{86}$$

rather than associate neutron-star phenomenology with a single preferred radius.

For the present discussion, the important point is how a new physical ingredient shifts the mass–radius relation. A change in the microscopic EOS can modify the stellar radius at fixed mass, while a modification of the gravitational field equations can produce a similar displacement even for the same EOS. The same is true for changes in the effective stress-energy source. Radius information is therefore most useful when combined with mass measurements and with observables that probe the stellar response in a different way.

7.3 Tidal deformability

Gravitational waves provide such a complementary probe through the tidal response of neutron stars in binary systems. During the inspiral, the external tidal field generated by one star induces a multipolar deformation in its companion. In GR, the dimensionless quadrupolar tidal deformability is

$$\Lambda = \frac{2}{3}k_2 \left(\frac{Rc^2}{GM} \right)^5 = \frac{2}{3}k_2 C^{-5}, \quad (87)$$

where k_2 is the quadrupolar tidal Love number and

$$C = \frac{GM}{Rc^2} \quad (88)$$

is the stellar compactness [33, 34, 41].

The strong C^{-5} dependence makes Λ particularly sensitive to the stellar radius. Even a moderate change in compactness can therefore produce a substantial change in the tidal response. This sensitivity is illustrated in Fig. 4 for fixed representative values of k_2 .

The binary neutron-star merger GW170817 provided the first gravitational-wave measurement with clear sensitivity to neutron-star tidal effects [23, 24]. The subsequent detection of GW190425 enlarged the binary-neutron-star sample, although its tidal information was considerably less constraining [26]. Joint analyses of gravitational-wave, NICER, pulsar-mass, and nuclear-theory information now provide stronger constraints on the dense-matter EOS than any of these channels alone [22, 27].

A commonly quoted reference quantity is the tidal deformability of a $1.4 M_\odot$ neutron star. For the purposes of the present discussion, the relevant scale remains

$$\Lambda_{1.4} \sim \mathcal{O}(10^2\text{--}10^3), \quad (89)$$

while the precise credible interval depends on the waveform model, priors, and assumptions used to parametrize the EOS.

An additional subtlety arises when the gravitational sector is modified. Equation (87) may still provide a useful definition of a dimensionless tidal response, but the value of k_2 cannot in general be taken from the corresponding GR solution. Modified gravity can change both the equilibrium background and the linear perturbation equations from which the Love number is obtained. A consistent comparison with gravitational-wave data therefore requires the stellar background and tidal perturbations to be calculated within the same gravitational framework.

7.4 Combining the observables

The main strength of neutron-star phenomenology comes from combining observables that respond differently to the microscopic EOS and the gravitational dynamics. Massive pulsars constrain the maximum mass, X-ray observations restrict the mass–radius relation,

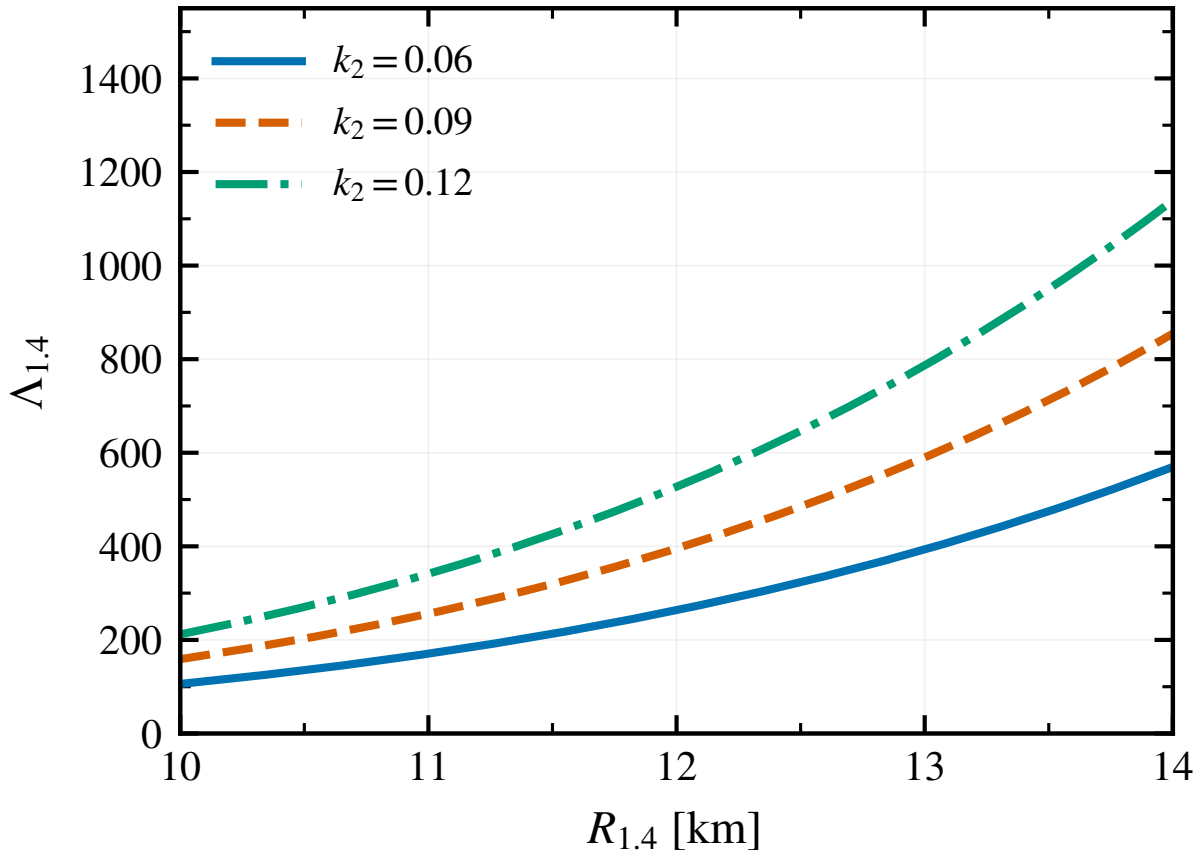


Figure 4: Dimensionless tidal deformability $\Lambda_{1.4}$ as a function of the radius $R_{1.4}$ for a $1.4 M_\odot$ neutron star and representative fixed values of the quadrupolar Love number k_2 . In GR, $\Lambda = (2/3)k_2 C^{-5}$, so the tidal response is strongly amplified by changes in stellar compactness. The curves are intended to illustrate this sensitivity and are not predictions of a particular EOS or quantum-gravity-inspired model.

and gravitational-wave measurements probe the tidal response. Modern multimessenger analyses combine these observations with theoretical information from nuclear physics to infer the EOS over an increasingly broad density range [22, 27].

A viable model must therefore reproduce several observables simultaneously, which can be summarized schematically as

$$\{M_{\text{max}}, R(M), \Lambda(M)\}_{\text{theory}} \sim \{M, R, \Lambda\}_{\text{obs}}, \quad (90)$$

where the symbol $\sim$ denotes consistency within the corresponding observational and theoretical uncertainties.

The need for a combined analysis follows from a central degeneracy in neutron-star physics. Changes in nuclear interactions, many-body correlations, particle composition, or high-density phase structure can shift M , R , and Λ in ways that resemble the effects of modified gravity or other new physics [5–7, 42]. A displaced mass–radius curve, for example, does not by itself reveal whether the origin of the shift lies in the EOS or in the gravitational sector.

This issue is particularly relevant for the three classes of models considered here. In the GUP realization of Sec. 4, the correction enters through the microscopic thermodynamics and therefore changes the EOS. In the noncommutative-inspired construction of Sec. 5, the effective spatial distribution and stress-energy structure of the source are modified. In the gravitational EFT and related scenarios of Sec. 6, the field equations and stellar equilibrium conditions may change directly. Despite their different microscopic interpretations, all three mechanisms ultimately feed into the same limited set of macroscopic observables.

Multimessenger observations should therefore not be expected to reveal a unique quantum-gravity signature in isolation. Their more realistic role is to reduce the region of parameter space in which a given quantum-gravity-inspired scenario remains compatible with neutron-star data. A meaningful constraint must be obtained together with the uncertainties of the dense-matter EOS and, when the gravitational sector is modified, with a self-consistent treatment of both the equilibrium configuration and its perturbations.

As measurements of neutron-star masses, radii, and tidal properties improve, this combined approach will become increasingly restrictive. Its value lies not in providing direct access to Planck-scale physics, but in testing how large an effective departure from the standard description of dense matter and gravity can remain compatible with the observed properties of neutron stars.

7.5 A minimum standard for a credible bound

The EOS–gravity degeneracy is not a small correction: variations of the high-density symmetry energy can exceed changes produced by some allowed alternative-gravity models [47]. A credible bound should state (i) the algebra, action, or source being constrained; (ii) the controlled expansion parameter; (iii) the thermodynamic and conservation equations; (iv) the EOS and composition range; (v) whether background and tides use the same theory; and (vi) the likelihoods and priors. Without these items, a limit may constrain a toy model but cannot be transferred to another GUP algebra, smearing prescription, or EFT coefficient.

8 Conclusions

Neutron stars occupy an unusual position in the search for physics beyond the standard descriptions of matter and gravity. They combine supranuclear densities, strong gravitational fields, and a growing set of precision observations, yet they remain enormously separated from the Planck regime. Their relevance to quantum-gravity phenomenology is therefore not that they probe the microscopic structure of spacetime directly, but that they provide a controlled astrophysical environment in which possible low-energy remnants of such physics can be tested.

The examples discussed in this article illustrate three distinct routes through which quantum-gravity-inspired effects can enter the neutron-star problem. Generalized uncertainty principles can modify the microscopic phase-space structure and consequently the

thermodynamics and equation of state. Noncommutative-inspired constructions provide a different possibility, in which the effective localization and stress-energy distribution of the matter source are altered. Higher-curvature corrections and scale-dependent gravitational couplings instead modify the gravitational dynamics and therefore the equations governing stellar equilibrium. This separation into matter-sector, effective-source, and gravitational corrections provides a useful framework for comparing models whose microscopic motivations may otherwise be very different.

A common conclusion emerges from all three cases. If the relevant corrections are genuinely controlled by the Planck scale and their dimensionless coefficients are of natural size, their effects under neutron-star conditions are expected to be extremely small. Observable changes in stellar structure generally require an additional ingredient: a lower effective scale, enhanced couplings, new low-energy degrees of freedom, or some other mechanism that amplifies the correction far below the Planck regime. Astrophysical bounds should therefore be interpreted primarily as constraints on the effective parameters of a particular phenomenological realization, rather than as direct measurements of a fundamental quantum-gravity scale.

The interpretation of such bounds is further complicated by the uncertainty of the high-density equation of state. Changes in nuclear interactions, particle composition, many-body correlations, or possible phase transitions can modify the mass–radius relation and tidal response in ways that resemble the effects of new gravitational or microscopic physics. No single neutron-star observable can therefore be expected to provide an unambiguous quantum-gravity signature. Reliable constraints require the dense-matter model, the effective source, and the gravitational dynamics to be treated consistently within the same stellar calculation.

This is where multimessenger observations become particularly important. Precise pulsar masses constrain the maximum mass that a viable model must support, X-ray observations probe the mass–radius relation, and gravitational-wave measurements provide complementary information through the tidal response. Their combined use can reduce degeneracies that remain unavoidable when any one observable is considered in isolation. The same consistency requirement applies on the theoretical side: when gravity is modified, both the equilibrium configuration and the corresponding perturbation equations must be derived within the same framework.

The most promising role of neutron stars in quantum-gravity phenomenology is therefore one of exclusion and consistency. Rather than providing direct access to Planck-scale physics, they can determine how large an effective departure from conventional dense-matter physics and General Relativity can remain compatible with observation. As mass and radius measurements improve, gravitational-wave observations accumulate, and the high-density equation of state becomes better constrained, this allowed parameter space will continue to shrink.

Future progress will depend as much on theoretical consistency as on observational precision. Meaningful tests require models in which the connection between the microscopic deformation, its effective low-energy scale, the stellar structure equations, and the predicted

observables is explicit. Establishing this connection is essential if neutron-star constraints are to be interpreted not merely as bounds on phenomenological parameters, but as useful information about the possible low-energy structure of a more fundamental theory of gravity.

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Data availability

No research datasets were generated or analyzed in this study. The numerical data and plotting scripts used for the illustrative figures are available from the author upon reasonable request.

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Conflict of interest

The author declares no conflicts of interest.

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